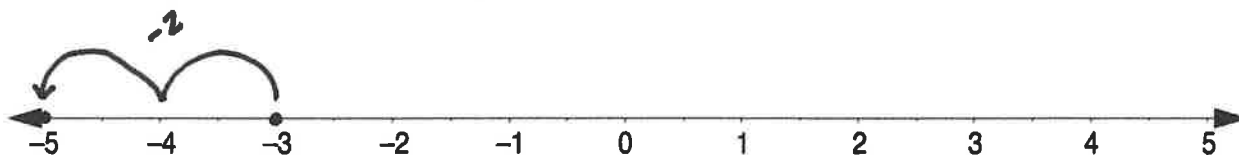


## Unit 3: Worked Examples

1. Use a number line to explain why the statement  $-5 < -3$  is true



2. Use the  $<$ ,  $>$  or  $=$  symbols to correctly compare each pair of values

a.  $0.4 > \frac{9}{40}$   
 $\downarrow$   $\downarrow$   
 $0.40$   $0.225$

b.  $-\frac{2}{5} < -\frac{5}{20}$   
 $\times 4$   $\downarrow$   $\downarrow$   
 $-\frac{8}{20}$   $-\frac{5}{20}$

c.  $|2.3| < |-5|$   
 $\downarrow$   $\downarrow$   
 $2.3$   $5$

3. Complete each statement

a.  $|-2.3| =$

$2.3$

b.  $|2.3| =$

$2.3$

c.  $|?| = 5\frac{1}{2}$

$5\frac{1}{2}$  or  $-5\frac{1}{2}$

4. A zoo veterinarian examined a tiger and determined the animal's weight to be 21.6 pounds below average. What value represents this situation? What is the absolute value of the of the tiger's weight loss?

$-21.6$

$|-21.6| = 21.6$

5. The captain of a fishing vessel is standing on the deck at 26 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 42 feet. Draw a model of this situation and use absolute values to determine the length of the rope.

